

CS 4720 HW 11

1ai) Over here, we have B and M as roots and they are parents of I as well and B, I and M are parents of G and G is the parent of J.

So, if we join all the factors then we get:

$$P(B, I, M, G, J) = P(B) P(M) P(I|B, M) P(G|I, B, M) P(J|G)$$

i) According to our observation, we can say that we must get $P(B, I, M) = P(B)P(M)P(I|B, M)$ but in this question we get $P(B, I, M) = P(B)P(I)P(M)$, which means that I is independent of B and M, but according to our observation it is not, hence it is not implied by the network.

ii) Since G is the only neighbor of J, we can say that every path from I to J must go through G, that is, a path from M must reach J in the following way:

$$I \rightarrow G \rightarrow J$$

So, we can say that G is non collider (since the pathways that go through G like this $I \rightarrow G \rightarrow J$) and part of the conditioning set (G, I), so we can say that these paths are blocked and there exists no way such that I can reach J, so adding it over here doesn't make any changes and hence (2) is asserted by the network.

Since G is the only neighbor of J, we can say that every path from M to J must go through G, that is, a path from M must reach J in the following way:

$$M \rightarrow I \rightarrow G \rightarrow J$$

So, we can say that G is non collider (since the pathways that go through G like this $I \rightarrow G \rightarrow J$) and part of the conditioning set (G, B, I), so we can say that these paths are blocked and there exists no way such that M can reach J, so adding it over here doesn't make any changes and hence (3) is asserted by the network.

b) In order compute the answer for this question, we must do the factorization of the Bayes net over here, so we get:

$$P(B, M, I, G, J) = P(B)P(M)P(I|B, M)P(G|B, I, M)P(J|G)$$

So, over here we get that:

$$P(b, i, \neg m, g, j) = P(b)P(\neg m)P(i|b, \neg m)P(g|b, i, \neg m)P(j|g)$$

Now, let us plug in the values from the question and multiply them so we get the following answer:

$$P(b,i,\neg m,g,j) = 0.9*0.9*0.5*0.8*0.9 = 0.292$$

Hence, the answer to this question is 0.292.

c) Over here, we can say that this is what the question demands from us:

$$P(J|B,I,M) = P(j|b,i,m)$$

Let us first use the Bayes net factorization over here, so we get that:

$$P(j|b,i,m) = P(j|g)P(g|b,i,m) + P(j|\neg g)P(\neg g|b,i,m)$$

So, after we plugin the values, we get that:

$$P(j|b,i,m) = 0.9*0.9 + 0*0.1 = 0.81$$

Hence, we can say that if someone breaks the law, and then indicted, and faces a highly motivated prosecutor, then there is a 81% probability that they will go to jail.

2) Over here, we can say that by looking at the burglary-alarm network, the CPTs can be assigned these values:

$$f_1(B) = P(B)$$

$$f_2(E) = P(E)$$

$$f_3(A,B,E) = P(A|B,E)$$

$$f_4(A,J) = P(J|A)$$

$$f_5(A,M) = P(M|A)$$

Now, if $J=t$, then $f_4(A, J) = P(J|A)$ becomes $f_4(A, J) = P(J = t|A)$ and if $M=t$, then $f_5(A, M) = P(M|A)$ becomes $f_5(A, M) = P(M=t|A)$, which basically removes J and M as variables over here.

Now, let us try and eliminate E (earthquake), where we just multiply all the factors that involve E , so we get:

$$f_2(E)f_3(A,B,E) = P(E = \text{true}) * P(A|B,E = \text{true}) + P(E = \text{false}) * P(A|B,E = \text{false}).$$

Now, let us eliminate the alarm variable A . In order to do that, we must multiply all the factors that involve A , so we get:

$$f_6(A,B)f_4(A,t)f_5(A,t)$$

And now, let us eliminate in our next step:

$$f_7(B) = f_6(A = \text{true}, B) * P(J = \text{true} \mid A = \text{true}) * P(M = \text{true} \mid A = \text{true}) + f_6(A = \text{false}, B) * P(J = \text{true} \mid A = \text{false}) * P(M = \text{true} \mid A = \text{false})$$

Now, since we have eliminated A (alarm) and E (earthquake), let us multiply all the remaining factors that are $f_1(B)$ and $f_7(B)$. So, now we get:

$$f_8(B) = f_1(B) * f_7(B)$$

Now, let us normalize everything to get our final answer.

$$P(B \mid J=t, M=t) = f_8(B) / (f_8(t) + f_8(f))$$

Hence, we can now calculate the posterior by looking at the standard values in the following way:

$$\begin{aligned} P(B = \text{true}, J = \text{true}, M = \text{true}) &= P(B = \text{true}) * [P(E = \text{true}) * P(A = \text{true} \mid B = \text{true}, E = \text{true}) * P(J = \text{true} \mid A = \text{true}) * P(M = \text{true} \mid A = \text{true}) + \\ &P(E = \text{true}) * P(A = \text{false} \mid B = \text{true}, E = \text{true}) * P(J = \text{true} \mid A = \text{false}) * P(M = \text{true} \mid A = \text{false}) + \\ &P(E = \text{false}) * P(A = \text{true} \mid B = \text{true}, E = \text{false}) * P(J = \text{true} \mid A = \text{true}) * P(M = \text{true} \mid A = \text{true}) + \\ &P(E = \text{false}) * P(A = \text{false} \mid B = \text{true}, E = \text{false}) * P(J = \text{true} \mid A = \text{false}) * P(M = \text{true} \mid A = \text{false})] \end{aligned}$$

Now, if we use the standard CPT values, we get:

$$P(B = \text{true}, J = \text{true}, M = \text{true}) = 0.0005922426 \text{ (approx)}$$

$$P(B = \text{false}, J = \text{true}, M = \text{true}) = 0.0014918576 \text{ (approx)}$$

Now, after normalizing everything, we get:

$$P(B = \text{true} \mid J = \text{true}, M = \text{true}) = 0.0005922426 / (0.0005922426 + 0.0014918576) = 0.284 \text{ (approx)}$$

Hence, the probability of having a burglary provided that we have both the calls is 0.284 or 28.4% approximately.

3a) Over here we have 2 evidence variables, that are Sprinkler and WetGrass that are assigned to true and then we have 2 non-evidence variables that are cloudy and rain. Since both of them are boolean values and hence we can say that the state space can have 4 (2x2) possible states over here.

b) Let us first try and state what all possible states we have right now:

- S1 = (C = true, R = true)
- S2 = (C = true, R = false)
- S3 = (C = false, R = true)

- $S4 = (C = \text{false}, R = \text{false})$

Since we have cloud and rain as our non-evidence variables, we can update both cloud and rain by probability $\frac{1}{2}$. Now, we have to compute the value of Q. Using the standard CPT values, we can assign values to the following variables:

$$P(\text{Cloudy} = \text{true}) = 0.5$$

$$P(\text{Rain} = \text{true} \mid \text{Cloudy} = \text{true}) = 0.8$$

$$P(\text{Rain} = \text{true} \mid \text{Cloudy} = \text{false}) = 0.2$$

$$P(\text{Sprinkler} = \text{true} \mid \text{Cloudy} = \text{true}) = 0.1$$

$$P(\text{Sprinkler} = \text{true} \mid \text{Cloudy} = \text{false}) = 0.5$$

$$P(\text{WetGrass} = \text{true} \mid \text{Sprinkler}, \text{Rain}) = (\text{already given in CPT})$$

Now, in order to fill the transition matrix, we must compute the conditional probabilities that are required by the Gibbs sampling. Since both Sprinkler and WetGrass are true the Markov blankets basically give us these values:

1) Resampling cloudy

$$P(C \mid S = \text{true}, R) \propto P(C) * P(S = \text{true} \mid C) * P(R \mid C)$$

a) $R = \text{true}$:

$$P(C = \text{true} \mid S, R) \propto 0.5 * 0.1 * 0.8 = 0.04$$

$$P(C = \text{false} \mid S, R) \propto 0.5 * 0.5 * 0.2 = 0.05$$

Normalize:

$$P(C = \text{true} \mid S = \text{true}, R = \text{true}) = 4/9 = 0.444$$

$$P(C = \text{false} \mid S, R) = 5/9 = 0.556$$

b) $R = \text{false}$

$$P(C = \text{true} \mid S = \text{true}, R = \text{false}) \propto 0.5 * 0.1 * 0.2 = 0.01$$

$$P(C = \text{false} \mid S = \text{true}, R = \text{false}) \propto 0.5 * 0.5 * 0.8 = 0.2$$

Normalize:

$$P(C = \text{true}) = 1/21 = 0.048 \text{ (approx)}$$

$$P(C = \text{false}) = 20/21 = 0.952$$

2) Resampling Rain

$$P(R \mid C, S = \text{true}, W = \text{true}) \propto P(R \mid C) * P(W = \text{true} \mid S = \text{true}, R)$$

a) $C = \text{true}$:

$$R = \text{true}: 0.8 * 0.99 = 0.792$$

$$R = \text{false}: 0.2 * 0.9 = 0.18$$

Normalize:

$$P(R = \text{true}) = 22/27 = 0.815$$

$$P(R = \text{false}) = 5/27 = 0.185$$

b) C = false:

$$R = \text{true}: 0.2 * 0.99 = 0.198$$

$$R = \text{false}: 0.8 * 0.9 = 0.72$$

Normalize:

$$P(R = \text{true}) = 11/51 = 0.216$$

$$P(R = \text{false}) = 40/51 = 0.784$$

Over here, eQ is computed in the following way:

$Q(s \rightarrow s') = \frac{1}{2} * P(\text{update } C \rightarrow \text{new } C) + \frac{1}{2} * P(\text{update } R \rightarrow \text{new } R)$ (Over here only 1 variable changes at a time)

Here are the final values:

	S1	S2	S3	S4
S1	$17/27 = 0.6296$	$5/54 = 0.0926$	$5/18 = 0.2778$	0.0
S2	$11/27 = 0.4074$	$22/189 = 0.1164$	0.00	$10/21 = 0.4762$
S3	$2/9 = 0.2222$	0.0	$59/153 = 0.3856$	$20/51 = 0.3922$
S4	0.0	$1/42 = 0.0238$	$11/102 = 0.1078$	$310/357 = 0.8683$

Note: Over here, each row sums up to 1 since it is a requirement for a Markov transition matrix.

c) Q^2 is basically the matrix of 2-step transition probabilities. The entry (i,j) basically tells us the probability that the Markov chain, which starts at state i, should be in state j after we get 2 Gibbs sampling steps.

d) We can say that over here as $n \rightarrow \infty$, Q^n basically converges to a matrix where every row has the same stationary distribution π , which basically means that no matter wherever we start, if we keep doing the Gibbs steps or apply Q, the distribution over here basically converges to π , also in the limits over here, the i^{th} row of Q^n is basically and equals the stationary distribution.

e) Let us first choose any starting distribution over the 4 Gibbs states, so let us call this initial distribution μ . Now, if we multiply by Q repeatedly then we get $\mu * Q^n$, for large n. So, over there as n gets large, the multiplication makes the distribution converge to the stationary distribution of the Gibbs sampler. After we do it a certain number of times, we get $\mu * Q^n = \pi$, which means that π is the long run probability of each state under Gibbs sampling, which means that it

does not depend on the starting point once n becomes larger. So, in order to answer this question, we will just add the probabilities where Rain = true, that is, $\pi(C=\text{true}, R=\text{true}) + \pi(C=\text{false}, R=\text{true})$, which is not a practical solution because this is basically a tiny example where we have just considered that Q is a 4×4 matrix, so in general for n variables we have $2^N \times 2^N$, which is an exponential size where it is impossible to store or compute for real networks, so over here we can basically say that using Q^n for inference is not practical, instead we can just run the Gibbs sampling and estimate probabilities from samples.